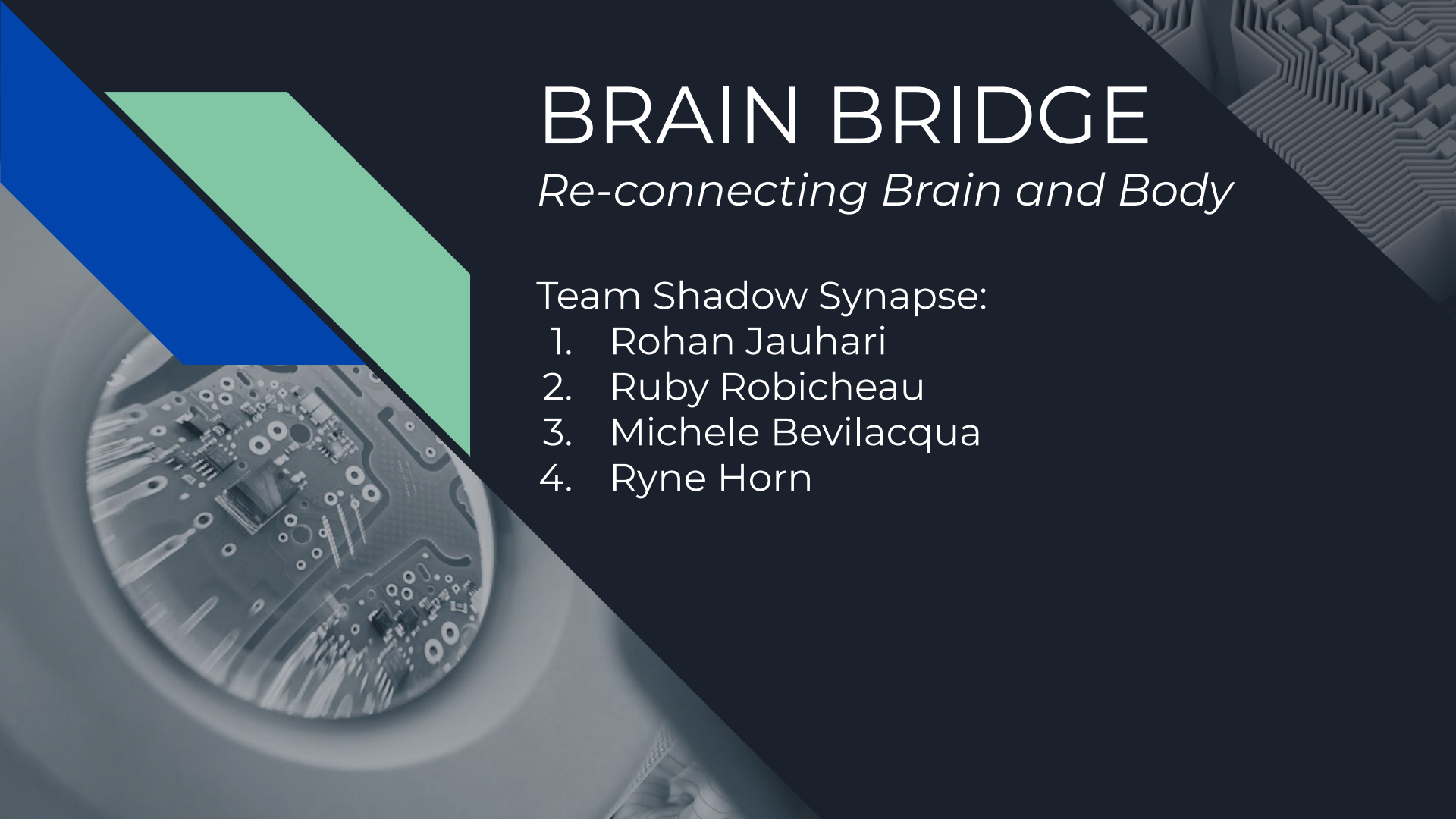




BRAIN BRIDGE

Re-connecting Brain and Body

Team Shadow Synapse:

1. Rohan Jauhari
2. Ruby Robicheau
3. Michele Bevilacqua
4. Ryne Horn

The Problem: **Impact of Paralysis**

- 15+ million people worldwide living with spinal cord injury (SCI)
 - 500,000 new cases annually
 - 28% of households with paralyzed individuals make <\$15,000/year
 - Lifetime economic burden: \$1.12-1.87 million per case
- 
- An abstract graphic on the right side of the slide, consisting of several overlapping, semi-transparent geometric shapes. A prominent green triangle points downwards, and a blue triangle points upwards, both appearing to be part of a larger, complex structure of dark grey and black polygons.



Current Technology Gaps

- Existing BCIs lack precision and adaptability
- Non-invasive systems require extensive setup (30+ min)
- Signal degradation requires frequent recalibration
- Limited translation to real-world functional independence



Solution

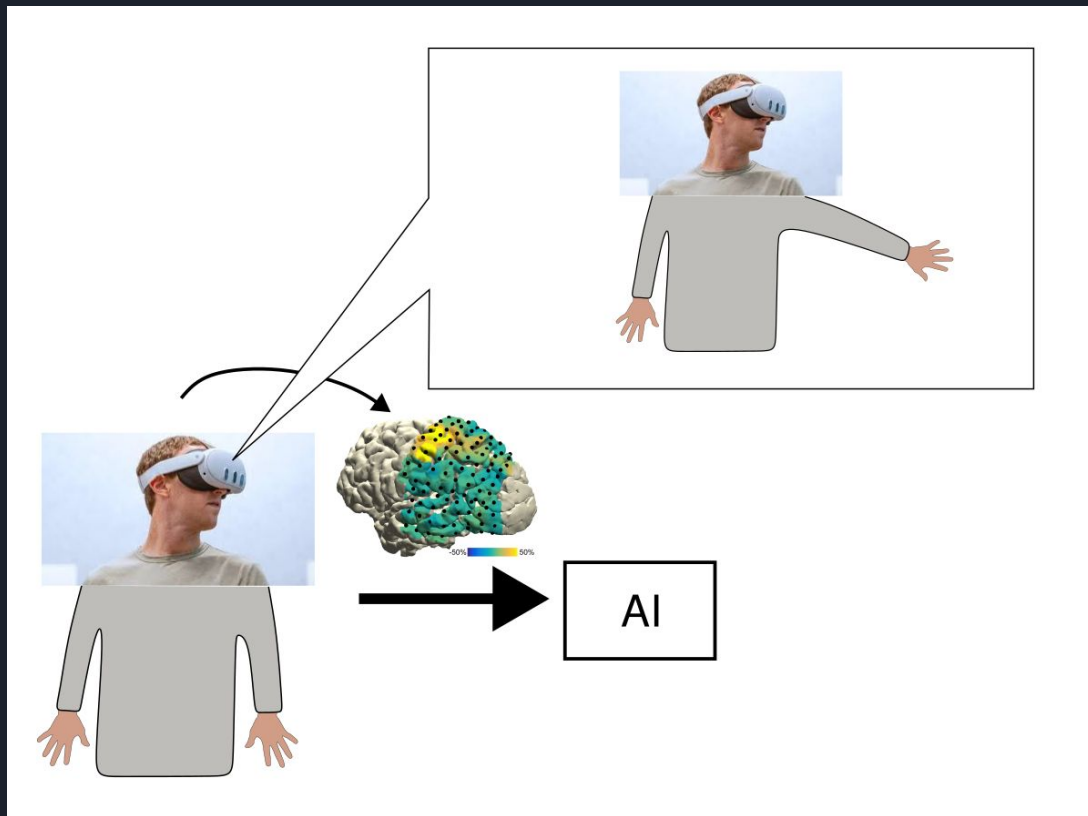
01 Neural Signal Acquisition:
High-density ECoG arrays over motor cortex

02 VR Training & Calibration Environment:
Immersive, task-specific training

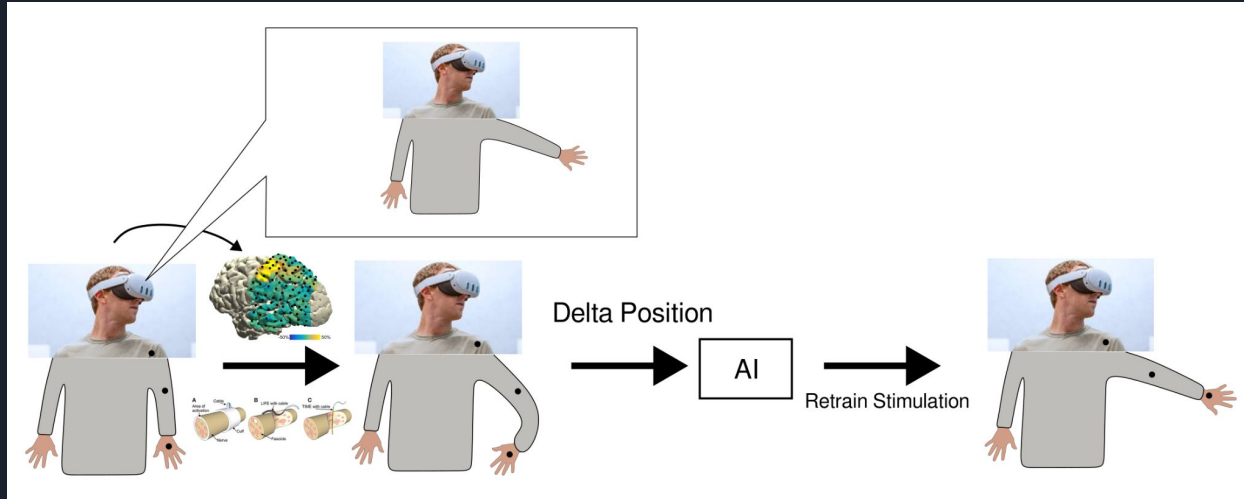
03 AI-Driven Neural-Muscular Mapping:
Adaptive deep learning algorithms

04 Precision Muscle Stimulation:
Coordinated multi-site muscle activation

Architecture - Training Phase



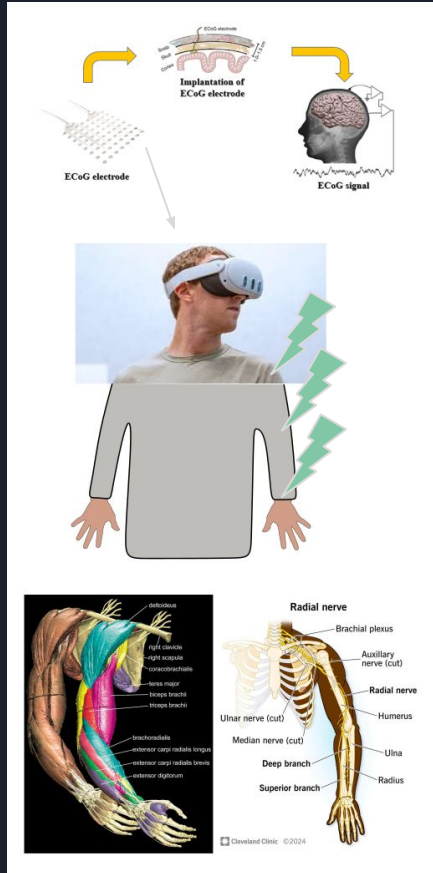
Architecture - Calibration Phase



Architecture - Usage Phase

The flow

1. User attempts an action
2. The brain implant perceives the corresponding signals
3. The implant transmits
4. Limb implant receives
5. Action





The Wow:

User Independence

Calibration without clinical assistance

Natural Movement

Fine-grained control beyond on/off functionality



VR Integration

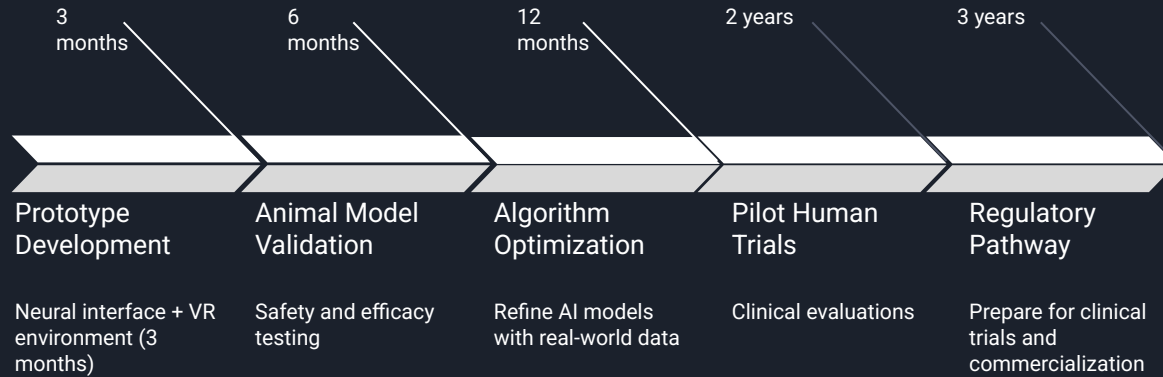
Immersive, engaging training environment

Adaptive Learning

Self-calibrating AI overcomes signal drift issues



Project timeline





THANK YOU